SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Operating Systems
COURSE CODE: CSA04

COURSE OUTCOME COVERED

CO3: Evaluate memory management mechanisms such as linking, paging, and segmentation to determine their effectiveness for different system requirements

Assessment Tool Weightage: 30%

QUESTIONS: 5 Total Marks: 50

CASE STUDY

1. An embedded system has limited physical memory (64 MB) and runs several small applications. Each application consists of multiple modules, and memory usage must be minimized. The system should load only the required modules when necessary.

   Tasks:

   a. Evaluate whether static linking or dynamic linking would be more suitable for this system.

   b. Compare the memory and storage requirements of the two approaches. c. Identify one disadvantage of your selected approach.

   d. Justify your final choice based on the requirements of the embedded system.

2. A cloud server runs hundreds of processes simultaneously. Processes have different memory requirements and may start or terminate at any time. The system should minimize external fragmentation and efficiently utilize available physical memory. **Tasks:**

   a. Evaluate contiguous memory allocation and paging for this system. b. Determine which mechanism would provide better memory utilization. c. Explain how the selected mechanism handles processes that require non contiguous memory.

   d. Justify your choice based on fragmentation, memory utilization, and process management.

3. A software application consists of several logically independent components such as:

   • Code

- Global data
- Stack
- Heap
- Shared libraries

The developers want memory protection between these components and want different access permissions for each logical section.

Tasks:

a. Evaluate whether paging or segmentation is more suitable for this application.

b. Explain how the selected technique can provide protection and controlled access.

c. Identify one limitation of the selected technique.

d. Justify your choice based on logical organization, protection, and memory utilization.

4. A modern operating system must support:

- Large applications
- Multiple processes
- Efficient physical memory utilization
- Protection between processes
- Virtual memory
- Programs larger than available physical memory

The system designers are considering paging, segmentation, and segmentation with paging.

Tasks:

a. Compare the three memory management mechanisms for the given requirements.

b. Evaluate which mechanism provides the best combination of protection, flexibility, and memory utilization.

c. Explain how your selected mechanism handles logical memory and physical memory.

d. Justify why the other two mechanisms are comparatively less suitable.

5. Consider two systems:

System A:

A high-performance gaming computer with large RAM, where applications require fast memory access and frequently load large executable files.

System B:

A multi-user server running many independent processes with varying memory requirements.

The OS designers are considering contiguous allocation, paging, and segmentation. **Tasks:**

a. Evaluate the suitability of each memory management technique for System A and System B.

b. Select the most appropriate mechanism for each system.

c. Compare your choices in terms of:

   a. Fragmentation
   b. Memory utilization
   c. Protection
   d. Address translation overhead
   e. Flexibility

d. Provide a final recommendation for both systems with appropriate

justification. **RUBRICS FOR CALCULATION**

Criteria	**Weightage**	**Level 4 – Exemplary (5 Marks)**	**Level 3 – Proficient (4 Marks)**	**Level 2 – Developing (2–3 Marks)**	**Level 1 – Beginning (0–1 Mark)**
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand